

The original full packet: exact formal boundary comparison

Split-Zero research continuation

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1 The original full packet at its formal boundary

This calculation concerns the original polynomial at its original marked parameter $t = 0$. It does not replace that polynomial by a single-primary polynomial. All its primary pieces occur simultaneously in the target of a proved map. The completion used below is in the deformation variable u ; its coefficient, differential, and marked fibre maps are specified explicitly.

1.1 The fixed polynomial and the two complete complexes

Fix the ordered nonempty finite set $Z = (\rho_1, \dots, \rho_r)$ of selected actual zeros of $g = 2\xi$, with their complete orders $m_\rho \geq 1$. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho} = s^d + \sum_{b=0}^{d-1} h_b s^b, \quad d = \sum_{\rho} m_\rho, \quad \Phi(s) = \frac{s^{d+1}}{d+1} + \sum_{b=0}^{d-1} \frac{h_b s^{b+1}}{b+1}. \quad (\text{FPB.1})$$

Thus $\Phi' = h$ and $\Phi(0) = 0$. In particular the constants $c_\rho := \Phi(\rho)$ are fixed by the original polynomial; they are not the constants of separately reassigned primitives.

Let $A = \mathbb{C}[[u]]$, $K = \mathbb{C}((u))$, and define

$$R = \mathbb{C}[s][[u]] = \varprojlim_n \mathbb{C}[s, u]/(u^n), \quad \widehat{\mathcal{C}} = [R \xrightarrow{L} R ds], \quad Lf = (u \partial_s f + h f) ds. \quad (\text{FPB.2})$$

Degrees are zero and one, respectively. A coefficient of each power of u is a polynomial in s , with no common degree bound required. Thus R is not being identified with $A[s]$. It is complete and separated for the u -adic topology. Its differential is continuous for that topology.

For each ρ , write $y = s - \rho$, $m = m_\rho$, and $N = m + 1$. Define the entire polynomial unit germ

$$a_\rho(y) = \frac{N(\Phi(\rho + y) - \Phi(\rho))}{y^N}, \quad a_\rho(0) = \prod_{\sigma \neq \rho} (\rho - \sigma)^{m_\sigma} \neq 0. \quad (\text{FPB.3})$$

The first equality is polynomial because $\Phi(\rho + y) - \Phi(\rho)$ vanishes to order N . For the second, integrate $h(\rho + y) = y^m \prod_{\sigma \neq \rho} (\rho - \sigma + y)^{m_\sigma}$ coefficient by coefficient. Choose and retain a specified number b_ρ with $b_\rho^N = a_\rho(0)$. The branch

$$A_\rho(y) = b_\rho \sum_{j \geq 0} \binom{1/N}{j} \left(\frac{a_\rho(y)}{a_\rho(0)} - 1 \right)^j, \quad w = y A_\rho(y), \quad y = \psi_\rho(w) \quad (\text{FPB.4})$$

is a formal coordinate with a uniquely specified formal inverse. Indeed its linear coefficient is $b_\rho \neq 0$; equating the coefficient of w^n in $\psi A_\rho(\psi) = w$ determines the coefficient of w^n in ψ uniquely by

division by b_ρ , after the preceding coefficients have been determined. The binomial identity gives $A_\rho^N = a_\rho$. Consequently

$$\Phi(\rho + \psi_\rho(w)) = c_\rho + \frac{w^N}{N}, \quad h(\rho + \psi_\rho(w))\psi'_\rho(w) = w^m. \quad (\text{FPB.5})$$

These are equalities of formal series retaining the original constant.

Put $R_\rho = \mathbb{C}[[w]][[u]]$, equipped with its u -adic topology and the coefficientwise formal derivative in w . Although its elements can also be written as double formal series, only the stated topology is used in the completion arguments. Set

$$\mathcal{C}_\rho = [R_\rho \xrightarrow{L_\rho} R_\rho dw], \quad L_\rho f = (u\partial_w f + w^m f) dw. \quad (\text{FPB.6})$$

The actual map of complexes is

$$F_\rho^0 f = f(\rho + \psi_\rho(w)), \quad F_\rho^1(f ds) = f(\rho + \psi_\rho(w))\psi'_\rho(w) dw, \quad F = \bigoplus_{\rho \in Z} F_\rho. \quad (\text{FPB.7})$$

Substitution is performed on each polynomial coefficient of u . The chain rule and (FPB.5) prove $F_\rho^1 L = L_\rho F_\rho^0$. The Jacobian in degree one is part of this identity. The map F is not asserted to be surjective on cochains.

1.2 Complete remainder, inverse, and chain homotopies

For $f = \sum_{n \geq 0} u^n f_n(s) \in R$, recursively divide

$$f_n - \partial_s q_{n-1} = h q_n + r_n, \quad q_{-1} = 0, \quad \deg r_n < d. \quad (\text{FPB.8})$$

Each division is the original monic polynomial division. Then $f = Lq + r$, where $q = \sum u^n q_n$, $r = \sum u^n r_n$, and the one-form symbol is understood. Conversely this identity determines every coefficient by (FPB.8). In particular L is injective, and

$$R = L(R) \oplus \mathbb{C}[s]_{<d}[[u]], \quad H^0(\widehat{\mathcal{C}}) = 0, \quad \mathcal{H} := H^1(\widehat{\mathcal{C}}) = \bigoplus_{b=0}^{d-1} A[s^b ds]. \quad (\text{FPB.9})$$

For injectivity one can also take the first nonzero coefficient of u in $Lq = 0$, obtaining $h q_n = 0$. Since $\mathbb{C}[s]$ is a domain, this is impossible.

In the local ring, divide every coefficient of $f_n - \partial_w q_{n-1} \in \mathbb{C}[[w]]$ uniquely by w^m :

$$f_n - \partial_w q_{n-1} = w^m q_n + r_n, \quad \deg r_n < m. \quad (\text{FPB.10})$$

The same coefficient recursion proves

$$H^0(\mathcal{C}_\rho) = 0, \quad \mathcal{H}_\rho := H^1(\mathcal{C}_\rho) = \bigoplus_{a=0}^{m_\rho-1} A[w^a dw]. \quad (\text{FPB.11})$$

No convergence of an infinite sum of fixed u -order is needed: each recurrence is a single formal division in w . The resulting quotient and remainder maps, denoted Q, R_h globally and $Q_\rho, R_\rho^{\text{rem}}$ locally, are continuous and A -linear. Their constructions agree with the polynomial-exponential

remainder when the input is a polynomial, because that remainder has the same unique decomposition. Thus (FPB.9) is also the completion of the original global lattice, rather than a differently presented quotient.

In local coordinates the remainder has an explicit coefficient formula. Define

$$p_{a,q} = \prod_{\ell=0}^{q-1} (a + 1 + \ell N), \quad p_{a,0} = 1.$$

For $0 \leq a < m$ and $q \geq 0$,

$$R_\rho^{\text{rem}}(w^{a+qN}) = (-u)^q p_{a,q} w^a, \quad R_\rho^{\text{rem}}(w^{m+qN}) = 0. \quad (\text{FPB.12})$$

To prove this, use $L_\rho(w^{j-m}) = w^j + u(j-m)w^{j-N}$ for $j \geq m$. The case $j = m$ gives zero remainder. Repeatedly reducing a monomial in the other congruence classes gives precisely the displayed product. For an infinite series in w , its coefficient of $u^q w^a$ uses just the coefficient of w^{a+qN} . For a series already containing u , the coefficient of any specified power of u uses a finite sum. This proves (FPB.12) at the stated complete types.

Write the matrix induced by F in the original ascending monomial columns and the ordered local rows (ρ, a) as $T(u)$. Then

$$T_{(\rho,a),b}(u) = \sum_{q \geq 0} (-u)^q p_{a,q} [y^{a+qN}] (\rho + y)^b a_\rho(y)^{-(a+qN+1)/N}, \quad (\text{FPB.13})$$

$$0 \leq b < d, \quad 0 \leq a < m_\rho.$$

Every fractional power uses the branch in (FPB.4), so its constant is $b_\rho^{-(a+qN+1)}$. For proof of this formula, formal change of variable in a residue gives, for any formal germ $f(\rho + y) \in \mathbb{C}[[y]]$,

$$[w^j] f(\rho + \psi(w)) \psi'(w) = [y^j] f(\rho + y) a_\rho(y)^{-(j+1)/N}. \quad (\text{FPB.14})$$

Indeed the left side is the coefficient of $w^{-1} dw$ in $f(\rho + \psi) \psi' w^{-j-1} dw$; substituting $w = y A_\rho(y)$ gives $f(\rho + y) y^{-j-1} A_\rho(y)^{-j-1} dy$. Formal residues are unchanged by this substitution: for monomials other than exponent -1 , this follows by differentiating a formal Laurent primitive; for exponent -1 , $w'/w = 1/y + A'_\rho/A_\rho$ has residue one. Combining (FPB.14) and (FPB.12) proves (FPB.13).

At $u = 0$, let $E_h = \mathbb{C}[s]/h$ and $E_\rho^w = \mathbb{C}[w]/w^{m_\rho}$. Then

$$T_0 : E_h ds \longrightarrow \bigoplus_{\rho} E_\rho^w dw, \quad [f] \longmapsto ([f(\rho + \psi_\rho) \psi'_\rho] dw)_\rho \quad (\text{FPB.15})$$

is invertible. Here is its explicit inverse. Given local remainders $q_\rho(w)$, set

$$f_\rho(y) = q_\rho(y A_\rho(y)) \frac{d}{dy} (y A_\rho(y)) \pmod{y^{m_\rho}}. \quad (\text{FPB.16})$$

Set $H_\rho = h/(s - \rho)^{m_\rho}$ and let $B_\rho(y)$ be the truncated inverse of $H_\rho(\rho + y)$ modulo y^{m_ρ} , calculated by the recursion for the inverse of a series with nonzero constant. The original Hermite idempotent is the degree-below- d remainder $e_\rho = \text{rem}_h(H_\rho B_\rho(s - \rho))$. The inverse of (FPB.15) is

$$(q_\rho)_\rho \longmapsto \text{rem}_h \sum_{\rho} e_\rho(s) f_\rho(s - \rho). \quad (\text{FPB.17})$$

Each e_ρ is one modulo its own primary factor and zero modulo every other one. The product of the distinct primary factors divides a polynomial divisible by each: repeatedly apply the Bezout identity to pairs of coprime factors. Thus their simultaneous jets determine a unique remainder modulo h . Finally $\psi'(w)w'(\psi(w)) = 1$ verifies (FPB.16) in both directions. This proves (FPB.17), without assuming the desired comparison.

With $T(u) = \sum_{n \geq 0} u^n T_n$, define

$$V_0 = T_0^{-1}, \quad V_n = -T_0^{-1} \sum_{i=1}^n T_i V_{n-i} \quad (n \geq 1). \quad (\text{FPB.18})$$

For $V(u) = \sum_{n \geq 0} u^n V_n$, it follows coefficient by coefficient that $TV = 1$. The same recursion or the adjugate formula over A , since $\det T_0 \neq 0$, gives $VT = 1$. Therefore

$$F_* : \mathcal{H} \xrightarrow[\sim]{T(u)} \bigoplus_{\rho} \mathcal{H}_{\rho} \quad (\text{FPB.19})$$

is the required isomorphism of complete original lattices.

There are explicit chain homotopies as well. Let i_h and i_{loc} include remainder representatives into degree one, and let p_h, p_{loc} denote their remainder projections, with zero maps in degree zero. The quotient maps Q, Q_{loc} are degree-minus-one maps and satisfy

$$1 - i_h p_h = dQ + Qd, \quad 1 - i_{\text{loc}} p_{\text{loc}} = dQ_{\text{loc}} + Q_{\text{loc}}d. \quad (\text{FPB.20})$$

Define $J = i_h T^{-1} p_{\text{loc}}$ as a map from the direct sum of local complexes to the global complex. Since remainder vanishes on every differential, this is a chain map. The identity $p_{\text{loc}} F = T p_h$ gives $JF = i_h p_h$. Moreover $p_{\text{loc}}(1 - FJ) = 0$, and hence

$$1 - JF = dQ + Qd, \quad 1 - FJ = d(Q_{\text{loc}}(1 - FJ)) + (Q_{\text{loc}}(1 - FJ))d. \quad (\text{FPB.21})$$

For the second identity, in degree one use zero remainder to write $(1 - FJ)f = L_{\text{loc}} Q_{\text{loc}}(1 - FJ)f$; in degree zero, $FJ = 0$ and $Q_{\text{loc}} L_{\text{loc}} = 1$. This also proves that these formulas retain the cochain information through a specific homotopy equivalence. The homotopies are A -linear; no assertion that the chosen remainders are horizontal is needed.

1.3 The exact connection and the constant determinant

Localize the displayed complete complexes at u . On the global degrees zero and one respectively the original connection is

$$\nabla^0 = \partial_u - \frac{\Phi(s)}{u^2} + \frac{1}{u}, \quad \nabla^1 = \partial_u - \frac{\Phi(s)}{u^2}. \quad (\text{FPB.22})$$

All operators are defined on $R[u^{-1}]$: each element has a finite lower bound on its powers of u , and differentiation preserves that property. Direct calculation gives $[u\partial_s + h, \partial_u] = -\partial_s$ and $[u\partial_s + h, -\Phi/u^2] = -h/u$, proving $\nabla^1 L = L \nabla^0$. Under (FPB.7) the phase becomes exactly $c_\rho + w^N/N$. The change of variables and its Jacobian are independent of u , so F intertwines both degree connections on these localized complexes.

In the unchanged global monomial basis, divide

$$\Phi(s)s^b = h(s)q_b(s) + r_b(s), \quad \deg r_b < d, \quad 0 \leq b < d.$$

Let C have columns r_b and B have columns q'_b . The derivative already has degree below d . Subtracting Lq_b gives the original connection matrix

$$G(u) = -C/u^2 + B/u. \quad (\text{FPB.23})$$

For the local column w^a , $(c_\rho + w^N/N)w^a = w^m w^{a+1}/N + c_\rho w^a$. The same reduction gives

$$D(u) = \bigoplus_{\rho \in Z} \text{diag}_{0 \leq a < m_\rho} \left(-\frac{c_\rho}{u^2} + \frac{a+1}{N_\rho u} \right), \quad T' + DT = TG. \quad (\text{FPB.24})$$

The second identity is equality of actual matrices over K , obtained by applying the proved chain intertwining to each original basis column. It is also an all-order identity satisfied by the coefficient formula (FPB.13).

The determinant is explicit and independent of u :

$$\det T(u) = \det T_0 = \left(\prod_{\rho \in Z} b_\rho^{-m_\rho(m_\rho+1)/2} \right) \prod_{i < j} (\rho_j - \rho_i)^{m_{\rho_i} m_{\rho_j}}. \quad (\text{FPB.25})$$

First consider T_0 . The ordinary jet matrix has entries $\binom{b}{a} \rho^{b-a}$. Its determinant is the confluent Vandermonde product in (FPB.25), with coefficient one and the displayed order. One elementary verification starts with the Vandermonde determinant at distinct variables $x_{i,a}$, whose determinant is $\prod_{(i,a) < (j,b)} (x_{j,b} - x_{i,a})$: it is alternating, divisible by every displayed difference, and has that leading coefficient by the identity permutation in the determinant. Apply successive divided differences within each ordered cluster and then set every variable in cluster i to ρ_i . The divided-difference row of order a becomes the Taylor-coefficient row $f^{(a)}(\rho_i)/a!$. The denominators cancel precisely the within-cluster Vandermonde factors; each cross-cluster difference becomes $\rho_j - \rho_i$. This proves the asserted coefficient and sign. The map of a local jet $y^a dy$ to $\psi(w)^a \psi'(w) dw$ is triangular with diagonal $b_\rho^{-(a+1)}$, proving the formula for $\det T_0$.

For independence of u , C is multiplication by Φ modulo h . On the ρ -primary quotient, $\Phi(\rho+y) = c_\rho + O(y^{m_\rho+1})$, so it is exactly scalar c_ρ , with multiplicity m_ρ . Hence $\text{Tr } C = \sum_\rho m_\rho c_\rho$. The quotient q_b has degree $b+1$ and leading coefficient $1/(d+1)$, so B is upper triangular with diagonal $(b+1)/(d+1)$ and trace $d/2$. Also $\sum_{a=0}^{m-1} (a+1)/(m+1) = m/2$. Thus $\text{Tr } G = \text{Tr } D$. Taking the trace of (FPB.24) after multiplication by T^{-1} gives $(\det T)' = 0$. In characteristic zero a formal series with zero derivative is constant, proving (FPB.25).

Replacing the chosen branch by $b_\rho \epsilon$, $\epsilon^{N_\rho} = 1$, replaces w by ϵw . Since $\epsilon^{a+qN+1} = \epsilon^{a+1}$, it replaces the local rows of T by exactly $\text{diag}(\epsilon^{-(a+1)})T_\rho$. This is the complete branch-change map; the branch is not suppressed.

1.4 The original arithmetic source and its Jacobian

Keep the original $\mathcal{B}, V, D = -x\partial_x, \Theta$, seed ϕ_* , and Euler inverse F_h , with $\mathcal{M}\Theta\phi_* = g$ and $\mathcal{M}F_h = g/h$. Explicitly, V consists of the even Schwartz functions on $\mathbb{R}$ satisfying $\phi(0) = \int_{\mathbb{R}} \phi = 0$, and $\mathcal{B}$ consists of smooth functions on $\mathbb{R}_{>0}$ with $\sup_{x>0} x^b |D^n F(x)| < \infty$ for every integer b and every $n \geq 0$. The maps are $\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx)$ and $\mathcal{M}F(s) = \int_0^\infty F(x) x^s dx/x$; $\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$. The original source maps are

$$\alpha_h(P) = P(D)\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \quad \Theta\alpha_h(P) = \mathcal{T}_h(hP), \quad J_h \mathcal{T}_h(P) = v_h[P], \quad v_h = [g/h] \in E_h^\times. \quad (\text{FPB.26})$$

These are the inherited maps, with their original spaces and norms; the coordinate calculation does not replace those spaces. In each local primary factor retain every coefficient

$$v_\rho(y) = \sum_{j=0}^{m_\rho-1} \frac{1}{j!} \left(\frac{d^j}{ds^j} \frac{g(s)}{h(s)} \right)_{s=\rho} y^j \pmod{y^{m_\rho}}. \quad (\text{FPB.27})$$

Its constant is nonzero because the selected order is complete. The ordinary marked arithmetic operator is $A_h = M_s$, and the ordinary full-unit operator is $U_h = M_{v_h}$. Under (FPB.15), their exact marked images are

$$T_0 A_h T_0^{-1} = \bigoplus_{\rho} M_{\rho+\psi_\rho(w)}, \quad T_0 U_h T_0^{-1} = \bigoplus_{\rho} M_{v_\rho(\psi_\rho(w))}. \quad (\text{FPB.28})$$

These multiplications are on $E_\rho^w dw$, hence truncated at the complete original order. To check them, compose the formulas for T_0 and its inverse; the two Jacobians cancel in a conjugation, while the full Taylor series remains.

Let $z_\rho = e_\rho(s)(s - \rho)^{m_\rho-1} \in E_h$. Then

$$T_0 z_\rho = b_\rho^{-m_\rho} w^{m_\rho-1} dw \quad \text{in the } \rho\text{-factor, zero elsewhere,} \quad T_0 U_h z_\rho = v_\rho(0) b_\rho^{-m_\rho} w^{m_\rho-1} dw. \quad (\text{FPB.29})$$

Indeed $\psi(w) = b_\rho^{-1} w + O(w^2)$ and $\psi'(w) = b_\rho^{-1} + O(w)$; all higher terms vanish at degree m_ρ . Both displayed terminal coefficients are nonzero.

Away from the marked fibre, extend the original matrices A_h, U_h over A in the unchanged global monomial basis. Their exact local operators are the explicitly computable matrices

$$\mathcal{A}(u) = T(u) A_h T(u)^{-1}, \quad \mathcal{U}(u) = T(u) U_h T(u)^{-1}. \quad (\text{FPB.30})$$

Equations (FPB.13) and (FPB.18) determine every coefficient of these operators. They are not identified with naive multiplication on a twisted de Rham quotient. The exact obstruction to that identification is visible already on representatives: for a polynomial v ,

$$vLq = L(vq) - uv'q, \quad R_h(vLq) = -uR_h(v'q). \quad (\text{FPB.31})$$

At $u = 0$ this defect vanishes. For an original representative p of degree below d , write $vp = hq + r$, $\deg r < d$. Then the full deformed remainder is

$$R_h(vp) = r - uR_h(q'), \quad TR_h(vp) = Tr - uTR_h(q'). \quad (\text{FPB.32})$$

This proves the precise comparison with coefficientwise multiplication by $[v]$ in E_h , including its derivative term. The same identity in the local complex has v' replaced by the derivative of the actual pulled-back coefficient $v(\rho + \psi(w))$. For $v = s$ and $\deg p < d$, the quotient q is constant, so the defect in (FPB.32) is zero on that section; (FPB.31) still proves why this is not a multiplication map on arbitrary cochains.

1.5 The precise attachment to the mixed support carriers

The marked comparison contains two different, explicitly related maps. The algebra map is

$$S_0 : E_h \xrightarrow{\sim} \prod_{\rho} E_\rho^w, \quad [f] \mapsto ([f(\rho + \psi_\rho)])_\rho. \quad (\text{FPB.33})$$

It is a unital algebra isomorphism: the same CRT proof applies, with the Jacobian omitted in both directions. Put $j = (\psi'_\rho(w))_\rho$, an invertible element of its target. Then the actual one-form map is $T_0 = M_j S_0$, and it satisfies

$$T_0(ax) = S_0(a)T_0(x) \quad (a, x \in E_h). \quad (\text{FPB.34})$$

Thus (S_0, T_0) is an exact isomorphism of a coefficient algebra and its rank-one module of marked one-forms. It is this pair that attaches to the mixed carriers. In general T_0 itself is not a unital algebra map: it sends $1 ds$ to $j dw$. If multiplication by an invertible j were multiplicative, its value on $1 \cdot 1$ would force $j = j^2$, hence $j = 1$.

For completeness, the split extension of a unital ring map $f : R \rightarrow R'$ is $G(f)(\tau) = \tau$ and $G(f)(r^\bullet) = f(r)^\bullet$. The operations are the original ones: τ is additive zero and multiplicatively absorbing; supported addition and multiplication are those of the ring. These formulas prove directly that $G(f)$ is a unital semiring map and that the supported zero $e = 0^\bullet$ stays distinct from τ . In particular $G(S_0)$ retains the original absolute-base point and the supported zero. The synchronized map

$$G\left(\prod_\rho E_\rho^w\right) \longrightarrow \prod_\rho G(E_\rho^w)$$

sends τ to the all- τ tuple and a supported tuple to the tuple with every coordinate supported. It does not introduce or delete the other independent support masks in the target.

More generally every original independent coefficient-slot mask and every original source-leg mask is retained as an external label. On the module attached to a fixed coefficient face, apply F , its remainder maps, and its homotopies coefficientwise; on the label use the identity. Inclusions, projections, and zero insertions of labelled slots commute with this operation by linearity. A newly inserted supported-zero slot remains supported-zero because its label is unchanged, while a missing slot remains missing. The same statement holds for a proper-source kernel carried as an additional labelled summand: the map acts on its coefficient module and never quotients out the summand. For multiplicative carrier arrows it is S_0 , not the Jacobian module map, that supplies the coefficient-ring arrow. On the deformed lattices the exact coefficient action is its original action transported by $T(u)$, as in (FPB.30); no unwarranted multiplicativity of the remainder projection is used.

1.6 All ordered tensor factors and formal character blocks

For $k \geq 1$, use the original ordered complex on $\mathbb{C}[s_1, \dots, s_k][[u]]$ with differential

$$\mathcal{D} = \sum_{i=1}^k (u \partial_{s_i} + h(s_i)) ds_i \wedge (-). \quad (\text{FPB.35})$$

This is the completed tensor product over A of the one-factor complexes: modulo u^n , the ordinary tensor product has polynomial coefficients in all the s_i ; taking the inverse limit gives precisely the displayed ring. The one-factor split homotopies extend to these completed tensor products. On a tensor summand with a contractible factor, apply its degree-minus-one homotopy with the sign from all preceding degrees; the two cross terms cancel in the tensor differential. Iterating leaves only the ordered top remainder summand. Thus its cohomology is concentrated in degree k , with basis $\prod_i s_i^{b_i} ds_1 \wedge \dots \wedge ds_k$, $0 \leq b_i < d$, and rank d^k .

Apply (FPB.7) independently in every ordered factor. For clarity, the local complex at a tuple $\rho \in Z^k$ is taken on $\mathbb{C}[[w_1, \dots, w_k]][[u]]$, with its u -adic topology and the differential $\sum_i (u \partial_{w_i} + w_i^{m_{\rho_i}}) dw_i \wedge (-)$. This ring is not identified with the mere u -adic completion of an algebraic tensor product of one-variable formal-series rings. Its cohomology is computed by successive one-variable

versions of (FPB.10), with coefficient ring the formal series in the remaining variables. Each division and derivative in one variable commutes with those in another. The same split tensor homotopies therefore leave the top remainder spanned over A by $\prod_i w_i^{a_i}$, $0 \leq a_i < m_{\rho_i}$, and no other cohomology. On an original monomial the separate reductions are the product of the separate one-variable reductions. The target is the direct sum over every tuple, and its top lattice map and phase are exactly

$$T_k(u) = T(u)^{\otimes k}, \quad ds_1 \wedge \cdots \wedge ds_k \mapsto \prod_i \psi'_{\rho_i}(w_i) dw_1 \wedge \cdots \wedge dw_k, \quad \sum_i \Phi(s_i) = C_{\rho} + \sum_i \frac{w_i^{N_{\rho_i}}}{N_{\rho_i}}, \quad C_{\rho} = \sum_i c_{\rho_i}. \quad (\text{FPB.36})$$

No permutation of the ordered one-forms was made, so no additional sign occurs. The target connection on the column indexed by $(\rho, \mathbf{a})$ is

$$\partial_u - \frac{C_{\rho}}{u^2} + \frac{\beta_{\rho, \mathbf{a}}}{u}, \quad \beta_{\rho, \mathbf{a}} = \sum_i \frac{a_i + 1}{N_{\rho_i}}, \quad 0 \leq a_i < m_{\rho_i}. \quad (\text{FPB.37})$$

Tensoring (FPB.24), or differentiating $T^{\otimes k}$ in each factor, proves the exact intertwining with the original tensor-sum connection. In degree j , the original chain connection retains the correction $(k - j)/u$. All tuples with the same value of C_{ρ} remain as distinct summands. Their direct sum is the full block with that exponential constant; no assertion that these sums are distinct enters the proof.

The original arithmetic tensor action is $A^{(k)} = \sum_i A_{h,i}$, with ordered full unit $U_h^{\otimes k}$. Under the complete map these become $T_k A^{(k)} T_k^{-1}$ and $T_k U_h^{\otimes k} T_k^{-1}$. At $u = 0$, the images are the actual multiplications by $\sum_i (\rho_i + \psi_{\rho_i}(w_i))$ and $\prod_i v_{\rho_i}(\psi_{\rho_i}(w_i))$, respectively. In particular

$$T_{k,0} U_h^{\otimes k} \bigotimes_i z_{\rho_i} = \left(\prod_i v_{\rho_i}(0) b_{\rho_i}^{-m_{\rho_i}} \right) \prod_i w_i^{m_{\rho_i}-1} dw_1 \wedge \cdots \wedge dw_k \neq 0. \quad (\text{FPB.38})$$

For the original cyclic terminal in a tuple, put $K_{\rho} = \sum_i (m_{\rho_i} - 1)$. The multinomial formula gives, in that actual primary corner,

$$\left(\sum_i (s_i - \rho_i) \right)^{K_{\rho}} = \frac{K_{\rho}!}{\prod_i (m_{\rho_i} - 1)!} \prod_i (s_i - \rho_i)^{m_{\rho_i}-1}. \quad (\text{FPB.39})$$

Every other monomial has an exponent exceeding one of the retained orders and vanishes. Its theta-weighted image under $T_{k,0}$ therefore has the coefficient in (FPB.38) multiplied by exactly the factorial ratio in (FPB.39). This coefficient is nonzero over the original complex numbers. No factorial, Taylor unit, or Jacobian has been absorbed into a new choice of terminal vector.

Here are the precise formal monodromy conventions. For one local factor put $\zeta_{\rho} = e^{2\pi i/N_{\rho}}$, $\beta_{\rho, a} = (a + 1)/N_{\rho}$, and

$$W_{\rho, j a} = \zeta_{\rho}^{j(a+1)} - 1 \quad (1 \leq j \leq m_{\rho}), \quad d_{\rho, a} = N_{\rho}^{\beta_{\rho, a}-1} e^{\pi i \beta_{\rho, a}} \Gamma(\beta_{\rho, a}). \quad (\text{FPB.40})$$

These are the original homogeneous-model ordered contour constants: the ray of argument $(\arg u + \pi)/N_{\rho} + 2\pi j/N_{\rho}$ gives $d_{\rho, a} u^{\beta_{\rho, a}} \zeta_{\rho}^{j(a+1)}$ after the positive radial substitution $r^{N_{\rho}}/(N_{\rho}|u|)$, and subtraction of the zeroth ray gives W_{ρ} . The columns are independent because a polynomial $\sum_{a=0}^{m_{\rho}-1} x_a (z^{a+1} - 1)$ vanishing at all nontrivial N -th roots also vanishes at 1 and has degree below N , so is zero. The gamma factors are nonzero since $0 < \beta < 1$.

Adjoin formal symbols $e^{c_\rho/u}$ and the fractional powers $u^{\beta_{\rho,a}}$, with the derivative specified by their displayed exponents. A full formal period matrix for the original global connection is

$$\Pi^{\text{form}}(u) = \left(\bigoplus_{\rho} e^{c_\rho/u} W_{\rho} \text{diag}_a(d_{\rho,a} u^{\beta_{\rho,a}}) \right) T(u). \quad (\text{FPB.41})$$

Equations (FPB.24) and (FPB.40) verify $(\Pi^{\text{form}})' = \Pi^{\text{form}} G$ directly. Its exact determinant, in the specified block and monomial orders, is

$$\det \Pi^{\text{form}} = \det T_0 \left(\prod_{\rho} \det W_{\rho} \prod_{a=0}^{m_{\rho}-1} d_{\rho,a} \right) u^{d/2} \exp \left(\frac{\sum_{\rho} m_{\rho} c_{\rho}}{u} \right). \quad (\text{FPB.42})$$

Formal continuation $\log u \mapsto \log u + 2\pi i$ fixes the integer-power matrix $T(u)$ and the exponential symbols. It therefore has period monodromy

$$M^{\text{form}} = \bigoplus_{\rho} W_{\rho} \text{diag}_a(\zeta_{\rho}^{a+1}) W_{\rho}^{-1}. \quad (\text{FPB.43})$$

For the differential module's horizontal coefficient columns the solutions of $(\partial_u + D)v = 0$ instead have factors $e^{-c_{\rho}/u} u^{-\beta_{\rho,a}}$. Thus their monodromy characters are the inverses of the period characters in (FPB.43). This records the dual convention explicitly.

The ordered tensor period matrix is $(\Pi^{\text{form}})^{\otimes k}$, and its period character on $(\boldsymbol{\rho}, \mathbf{a})$ is $\exp(2\pi i \beta_{\boldsymbol{\rho}, \mathbf{a}})$. These matrices are diagonalizable. On the formal regular factors the unipotent logarithm is zero, and the dimension of the character-one space inside the block with constant C is exactly

$$\# \left\{ (\boldsymbol{\rho}, \mathbf{a}) : \sum_i c_{\rho_i} = C, \quad \sum_i \frac{a_i + 1}{N_{\rho_i}} \in \mathbb{Z}, \quad 0 \leq a_i < m_{\rho_i} \right\}. \quad (\text{FPB.44})$$

Repeated equal characters contribute their full multiplicity because all their ordered summands were retained. Formula (FPB.44) is about the formal regular character factors of this proved decomposition; it does not identify actual analytic inertia invariants or finite-field boundary stalks.

1.7 Horizontal formal projectors and the unchanged source form

Let P_{ρ} be the constant block projector on the ρ -summand of the local direct sum. The actual formal projector in the original global lattice is

$$\mathcal{E}_{\rho}(u) = T(u)^{-1} P_{\rho} T(u), \quad \mathcal{E}_{\rho}(0) = M_{e_{\rho}}, \quad \mathcal{E}'_{\rho} + [G, \mathcal{E}_{\rho}] = 0. \quad (\text{FPB.45})$$

The first identity defines all coefficients by (FPB.13) and (FPB.18). The marked identity follows from the weighted CRT map, whose Jacobian preserves each primary summand. Orthogonality, idempotence, and sum one follow by conjugating the corresponding identities for P_{ρ} . Differentiating the definition and using $T' = TG - DT$, with $[D, P_{\rho}] = 0$, gives the displayed horizontal identity. Thus the retained primary labels have explicitly horizontal projectors over the original complete lattice, including when critical values coincide; they have not merely been recovered from a dimension count.

There is also a marked constraint on equal-critical-value off-diagonal blocks. If $\rho \neq \sigma$ and $c_{\rho} = c_{\sigma} = c$, then

$$M_{e_{\sigma}} B M_{e_{\rho}} = 0 \quad \text{on } E_h. \quad (\text{FPB.46})$$

Take the degree-below- d representative f of an element supported on the ρ -factor. Multiplication by Φ modulo h is cf , so the original quotient defining Bf is exactly $q = (\Phi - c)f/h$, and $Bf = q'$. At σ , f vanishes to order at least m_σ , while $\Phi - c$ vanishes to order at least $m_\sigma + 1$. Therefore q vanishes to order at least $m_\sigma + 1$, and q' vanishes to order at least m_σ , proving (FPB.46). This proves the exact compressed first-order relation without assuming that all original critical values are different.

The original source form on the polynomial coefficient section is

$$(G_\theta)_{ab} = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{(1/2 + it)}^a (1/2 + it)^b \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 dt, \quad 0 \leq a, b < d. \quad (\text{FPB.47})$$

At a removable zero this integrand uses the entire quotient g/h . To verify its source type, put $x = e^z$. The original $L^2(dx)$ norm of $\mathcal{T}_h(P)$ becomes the $L^2(dz)$ norm of $e^{z/2} \mathcal{T}_h(P)(e^z)$. Its Fourier transform with the positive exponential convention is $P(1/2 + it)(g/h)(1/2 + it)$. Fourier Plancherel, with inverse factor $1/(2\pi)$, gives (FPB.47). The original $\mathcal{B}$ bounds imply square integrability and decay of all needed derivatives, so the displayed integrals are finite. If the norm is zero, the continuous original source function is zero; its Mellin transform $P(g/h)$ is zero. Since g/h is a nonzero entire function, a polynomial P with that property is zero. Hence G_θ is positive definite.

On a phase-coordinate vector representing the polynomial section $q = T_0 p$, the very same form is

$$G_{\text{phase}} = T_0^{-*} G_\theta T_0^{-1}. \quad (\text{FPB.48})$$

On the actual theta jet $q_\theta = T_0 U_h p$, the same source form instead has matrix

$$G_{\text{theta jet}} = (T_0 U_h)^{-*} G_\theta (T_0 U_h)^{-1}. \quad (\text{FPB.49})$$

Here $X^{-*} = (X^{-1})^*$. Substitution in the two quadratic forms proves both identities, including the full unit and every Jacobian factor. Their ordered tensor versions use the k -fold Kronecker products of the displayed maps and G_θ . There is no assertion that a possibly divergent formal $T(u)$ has a positive analytic Gram matrix at a nonzero numerical u .

The distinction is exact at the map level. Equations (FPB.7), (FPB.19), and (FPB.24) give a chain homotopy equivalence of the specified u -adic complexes and an isomorphism of their localized formal differential modules. No convergence of $T(u)$, no map from its formal local model rays to the original global rapid-decay cycles, and no Stokes transition matrices has been proved here. Consequently (FPB.41) is the specified formal period solution, whose relation to the original global connection is the proved identity $(\Pi^{\text{form}})' = \Pi^{\text{form}} G$; it is not declared to be an evaluated global analytic contour integral. The full original theta operator is related to it by (FPB.26)–(FPB.32) and their ordered tensors, rather than by an assumed Frobenius or weight intertwiner. In particular no arithmetic purity or Riemann-hypothesis conclusion is inserted into the boundary comparison.